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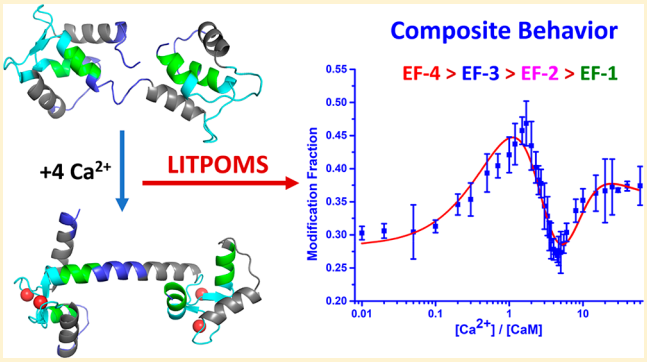
A Single Approach Reveals the Composite Conformational Changes, Order of Binding, and Affinities for Calcium Binding to Calmodulin

Xiaoran Roger Liu, Mengru Mira Zhang, Don L. Rempel, and Michael L. Gross*

Department of Chemistry, Washington University in St. Louis, One Brookings Drive, St. Louis, Missouri 63130, United States

Supporting Information

ABSTRACT: We found that a newly developed method named LITPOMS (ligand titration, fast photochemical oxidation of proteins and mass spectrometry) can characterize section-by-section of a protein the conformational changes induced by metal-ion binding. Peptide-level LITPOMS applied to Ca^{2+} binding to calmodulin reveals binding order and site-specific affinity, providing new insights on the behavior of proteins upon binding Ca^{2+}. We established that EF hand-4 (EF-4) binds calcium first, followed by EF-3, EF-2, and EF-1 and determined the four affinity constants by modeling the extent-of-modification curves. We also found positive cooperativity between EF-4, EF-3 and EF-2, EF-1 and allostery involving the four EF-hands. LITPOMS recapitulates via one approach the calcium-calmodulin binding that required decades of sophisticated development to afford versatility, comprehensiveness, and outstanding spatial resolution.



Our goal is a single approach to understand the complexities of protein−metal binding, especially multiple binding.[1,2] To do this, we chose calmodulin (CaM), a small calcium (Ca^{2+})-binding protein in all eukaryotic cells, that plays essential roles in Ca^{2+} signaling,[3,4] regulating cell motility, growth, proliferation, and apoptosis.[5−7] The protein has two homologous globular domains (N- and C-terminal) connected by a flexible linker.[8−11] Each domain contains a pair of Ca^{2+}-binding motifs called EF-hands and binds two Ca^{2+} ions cooperatively (CaM/Ca^{2+} stoichiometry is 1:4).[12,13] Subsequent protease susceptibility studies revealed cooperativity between and N- and C-terminal domains.[14] Each EF-hand binds with micromolar affinity, making the protein sensitive to intracellular Ca^{2+} concentration changes and responsive as a signaling protein. Each canonical EF-hand has a "helix−loop−helix" configuration, where a calcium loop sits between the two helices. Ca^{2+} binding alters interhelical angles in the EF-hands, moving them from "closed" to "open",[12] exposing hydrophobic sites that bind, and activating many target proteins.[7]

Understanding these interactions has required decades of research, during which various approaches were developed[2,13−26] and applied.[13,14,19−23,25] Fluorescence-based methods reveal overall conformational changes, Ca^{2+} binding affinities,[13,23] and, in combination with microfluidic mixers, the kinetics of conformational change upon binding.[21] NMR shows the EF-hand local conformational change induced by binding.[19] On the one hand, isothermal titration calorimetry examines the system thermodynamically, providing valuable binding constants.[20,22] On the other hand, the needs for specific fluorophore labeling, special sample preparation, and high sample quantity limit applications. Thus, a new approach that overcomes these problems and delivers improved spatial resolutions will be a significant improvement in tool set for protein−metal-ion binding.

Currently, NMR resolves protein solution-state structure with the highest spatial resolution. A titration-type affinity measurement for metal-ion binding, however, requires low micromolar concentrations, which are not accessible today to NMR. A mass spectrometry (MS) structural proteomics approach should significantly lower sample amount and eliminate the need for specific isotopic labeling as for NMR. Moreover, a structural proteomics approach may also greatly elevate the spatial resolution and sensitivity when compared with fluorescence.[24,25] Indeed, spatial resolution is achievable by protease digestion and MS.[27]

Recently, we developed ligand titration, fast photochemical oxidation of proteins and mass spectrometry (LITPOMS)[26] that combines ligand titration with footprinting by fast photochemical oxidation of proteins (FPOP).[28−30] We demonstrated its efficacy for 1:1 binding, where the results are similar to those of HDX.[25] That is, LITPOMS distinguishes binding from nonbonding, accurately reports the binding affinity, and reveals some of the critical binding residues.[26]

For more complicated signaling proteins, such as CaM, HDX in a titration format also shows binding and provides hints of remote conformational changes and allostery. But

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these proteins undergo complex conformational changes encompassing the whole protein, where some regions tighten and bind, others open and provide cooperativity and allostery. To meet the challenges of tracking these changes, we extended FPOP with its high footprinting speed and irreversible labeling to capture, section-by-section, this complexity and thereby reveal features not seen by HDX or specific amino-acid labeling.[25,31,32]

Here, we demonstrate that application of LITPOMS can meet the challenge and determine binding sites, binding affinities, binding orders, site-specific affinities, and importantly the complex, composite behavior of the protein upon metal ion binding. Moreover, it can reveal cooperativity and allostery. Information from LITPOMS recapitulates the long-term cumulative efforts to understand calcium−calmodulin binding.

In the current experiment, 1 μM calcium-free calmodulin (Apo-CaM) was first equilibrated with calcium at [Ca^{2+}]-to-[Apo-CaM] from 0 to 60, and the proteins were footprinted by FPOP.[26,34] Global response was recorded by MS under denaturing conditions. For peptide-level responses, the complexes were digested with trypsin/Lys-C, and the digested peptides were submitted to liquid chromatography (LC)/MS^2 analysis.[34] Integration of extracted ion chromatograms taken by LC-MS^2 allowed calculation of the modification fraction.

To ensure good coverage and high signal-to-noise ratio (S/N), an MS^2 method with a precursor-ion inclusion list was used for signal acquisition. The inclusion list was developed based on several test runs. To meet standards of resolvable extracted ion chromatograms, accurate precursor/fragment masses, and confident MS^2, eight distinct tryptic peptides were chosen to afford 99% sequence coverage (Figure 1, Figure S3,

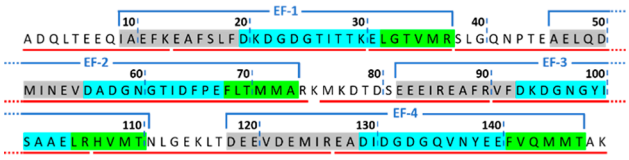


Figure 1. A 99% sequence coverage map of calmodulin showing regions of conformational change.[11] Red lines under the sequence indicate the observed peptides. The four EF-hands are superscored by blue brackets. Residues shaded with gray, cyan, and green represent the E helix, calcium loop, and F helix of each EF-hand, respectively.

and detailed description in Supporting Information). Like many Ca^{2+}-binding proteins, CaM contains four EF-hands consisting of an E helix, calcium loop, and F helix (peptides representing these regions are in gray, cyan, and green in Figure 1, respectively).

When titrated with Ca^{2+} by using the LITPOMS protocol, CaM shows four remarkable and distinct classes of behavior upon titration with Ca^{2+} (Figure 2), each successfully fit for binding. These curves are sensitive representations of opposing conformational changes that impart a composite of protection and deprotection (binding with tightening of structure vs loosening or opening in allostery).

The first class of behavior is the loss of protection in the linker region where the well-known transformation from compact to "dumbbell-like" occurs.[7,8,10,34] The experimental observations recapitulate the loosening of structure for region 76−90 (Class I behavior), representing the central linker between the two lobes together with part of helix E in EF-3, and for the region 107−126 connecting EF-3 and EF-4.

Class II behavior is for peptides 1−13 and 31−37 (black in Figure 2a,c). Modification extents for these peptides remain relatively constant as Ca^{2+} binds, indicating no involvement in binding and no remote conformational changes. Indeed, peptide 1−13 is part of the N-terminal of calmodulin, whereas peptide 31−37 is helix F in EF-1.

Class III behavior shows classical binding behavior, which is illustrated by peptide 14−30 that covers the helix E and calcium loop of EF-1 (magenta in Figure 2b). The modification fraction stays constant early in the titration, but when [Ca^{2+}]/[CaM] ≅ 2, this region becomes protected and shows a decrease in modification. Once saturated with calcium, peptide 14−30 remains protected at a modification fraction of ∼0.028.

The three other binding sites exhibit different behavior, termed Class IV. This behavior is composite, showing a combination of binding and conformational changes induced remotely (allostery).[14,21] Peptides 38−74, 91−106, and 127−148 join this category (blue panels in Figure 2d,f,h). Peptide 127−148 contains the calcium loop and F helix in EF-4, whose response shows a gain then a loss in protection when adding Ca^{2+}. The initial gain is due to calcium binding in EF-4, whereby residues in the calcium loop chelate Ca^{2+}, leading to increasing compactness and a decrease in solvent accessibility. When it passes the critical point of [Ca^{2+}]/[CaM] ≅ 1.5, the calcium loop in EF-4 becomes saturated with calcium, whereas the F loop becomes more exposed owing to binding-induced conformational change. Loss of protection of the F loop contributes to an increase of the modification fraction for peptide 127−148 at the later stage of the calcium titration. The modification fraction of peptide 91−106 stays relatively stable until [Ca^{2+}]/[CaM] ≅ 0.7, after which a decrease in modification occurs. Most of the calcium loop in EF-3 (residues 93−104) resides in this peptide, explaining the decrease of the modification fraction. Like peptide 127−148, an increase in modification fraction occurs at the later stage of the titration, owing to a binding-induced conformational change that facilitates calcium binding of other EF hands.

While falling in the class IV behavior, peptide 38−74 differs from its peers. With the addition of calcium, region 38−74 becomes less protected, seen as an increase in modification fraction. This increase peaks at [Ca^{2+}]/[CaM] ≅ 1.7, where EF-4 finishes binding with calcium, and EF-3 is on its way. Peptide 38−74 covers the linker between EF-1 and EF-2 as well as the whole EF-2 (helix E, calcium loop, and helix F). The observed loss in protection at the early stage of the titration is followed by a binding-induced conformational change. To be specific, binding of calcium by EF-3 and EF-4 facilitates a conformational change in EF-2, preparing it to take its own calcium. After [Ca^{2+}]/[CaM] ≅ 1.7, peptide 38−74 becomes protected, indicating that EF-2 starts to bind calcium. This discloses the allosteric behavior of calcium binding, where EF-2 is prepared to take calcium upon calcium binding at remote sites. Similarly, after adding 5 equivalents of calcium, peptide 38−74 becomes less protected, along with structural transitions for most of the protein (peptides 38−74, 76−90, 91−106, 107−126, and 127−148) to more extended conformations. During the later stage of the titration (i.e., [Ca^{2+}]/[CaM] ≥ 5), EF-1 is taking up calcium and transitioning into a calcium-bound state, while most of the protein is opening. The conformation of the holo-calmodulin with four bound calcium ions stabilizes when [Ca^{2+}] > 15

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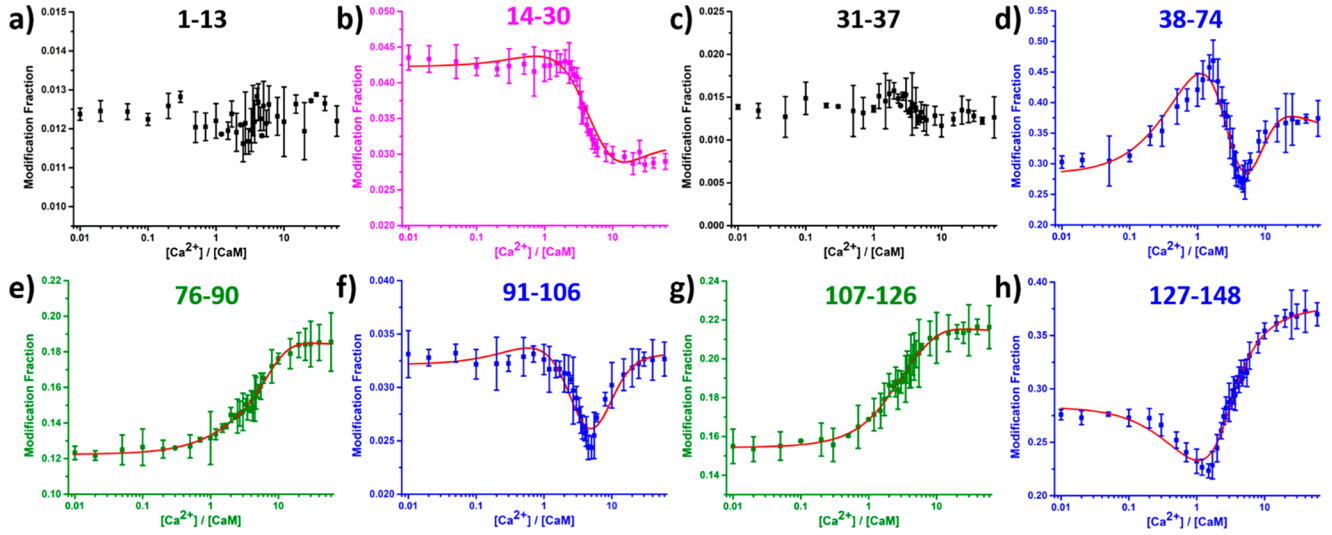


Figure 2. LITPOMS response at peptide level, where modification fractions were plotted as a function of calcium/calmodulin ratio. Four different classes of behaviors are shown in black (a, c), magenta (b), blue (d, f, h), and olive (e, g). Red solid lines in (b, d–h) are from fitting using an algorithm reported previously.[25,33] Data points represent average of two runs, and error bars are standard deviations. Modification fractions reported here account for all resolvable oxidations, for which a maximum of +32 were observed at peptide level.

equiv of the protein as seen by a nearly invariant modification extent.

Allostery upon Ca^{2+} binding has been difficult to assess. For example, proteolytic footprinting shows that, upon titration with Ca^{2+}, the susceptibility profiles of residues E31 and R37 of a truncated CaM (N-domain) differ from that of the N-terminus of the full protein.[14] This early insight into allosteric behavior of CaM upon Ca^{2+} binding was followed by protein grafting, revealing that the intrinsic Ca^{2+} affinities of isolated calcium loops in each individual EF-hand differ from those of the EF-hands in intact CaM. Although the differences reflect structural effects of adjacent residues and allostery,[35] neither study shows the actual behavior. LITPOMS greatly advances the understanding by revealing the overall calcium-binding dynamics of CaM as a function of calcium concentration, which, until now, has not been understood with such spatial detail.

In addition to following the complex dynamics upon Ca^{2+} binding, the results permit a determination of the order of Ca^{2+} binding (Figure 3). Given that bond formation increases protection (decreases reactivity) at a binding site, the onsets for decreases in modification should reveal the starting points for Ca^{2+} binding at each EF-hand. The onsets clearly show that the first binding occurs at EF-4 (peptide 127–148) and the last at EF-1 (peptide 14–30). The order of binding for EF-3 (peptide 91–106) and EF-2 (peptide 38–74) is ambiguous. Most of the decrease in modification fraction of EF-3 lags that of EF-2, indicating that the two binding events are highly competitive. Considering the earlier onset for EF-3, however, we conclude that this is the second site. Thus, the binding order is assigned as EF-4 > EF-3 > EF-2 > EF-1, which not only agrees with some previous conclusions that the C-terminal lobe binds first,[13] but also advances that picture by revealing the order within each lobe.

Following an assignment of binding order, we obtained four binding affinities for each EF hand (summary in Table 1) by fitting the various binding curves in Figure 2 by using a previously reported fitting algorithm,[25,33] where EF-1 and EF-2 were grouped into N-terminal lobe and where EF-3 and EF-4 were grouped in the C-terminal lobe. Positive cooperativity was assumed in each lobe, whereas the two lobes bind calcium independently.[13] Each state (apo-CaM, CaM-1-Ca^{2+}, CaM-2-Ca^{2+}, ...) has its distinct modification fraction. A search utilizes the experimental data to determine the system composition at

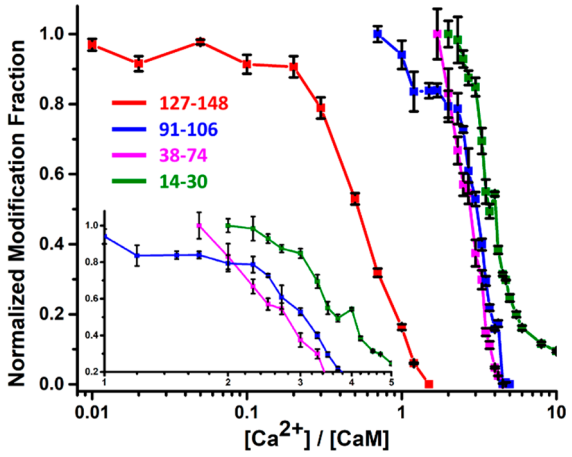


Figure 3. Normalized (to 1) LITPOMS responses for selected peptides showing increases in protection (decreases in modification) for four peptides representing the four EF hands of CaM. (inset) Expanded decreases in modification for peptides 91–106, 38–74, and 14–30. Data points are averages of two independent runs, and error bars are standard deviations.

Table 1. Summary of Binding Order and Site-Specific Binding Affinities for Calcium–Calmodulin System

binding order	EF-hand	LITPOMS K_i (M^{-1})	literature[13] K_i (M^{-1})
1	EF-4	1.4×10^6	8.0×10^4
2	EF-3	6.2×10^6	4.0×10^6
	C-term lobe	8.6×10^{12}	3.2×10^{11}
3	EF-2	4.1×10^4	2.5×10^4
4	EF-1	2.9×10^6	4.0×10^5
	N-term lobe	1.2×10^{11}	1.0×10^{10}

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different calcium concentrations, after which binding affinities and a peptide-specific fitting curve are obtained.

The results (Table 1) show that LITPOMS successfully reveals the binding order and the site-specific binding affinities for four EF-hands in calmodulin upon binding with calcium. The binding affinities for EF-2 and EF-3 agree reasonably well with those in the literature (within a factor of 1.6), whereas for EF-1 and EF-4 they are within 20-fold.[13] Although an advantage of LITPOMS for affinity measurements may be the peptide-level modeling is of at least four binding curves rather than a single global-level fluorescence response,[13] we view the affinities as estimates that need further validation. Nevertheless, the affinity constants show that the binding in each lobe is positively cooperative (EF-3 > EF-4 and EF-1 > EF-2).

In summary, we find that LITPOMS reveals the composite behavior for regions covering all of calmodulin upon binding with calcium and shows clearly the allosteric changes in, for example, region 38−74 caused by the binding. The composite behavior at peptide level is due, in part, to the convolution from the long peptide (e.g., 38−74), which can potentially be dissected through enzymatic digestion by a different protease or by tracking specific residues in that peptide. Further, the outcome illustrates the cooperativity within N-terminal and C-terminal lobes but also between the two lobes as demonstrated by the behavior of peptide 38−74 at early stage of titration. In addition to this detail, we can also determine the binding sites, the binding order, and site-specific binding affinities, which agree reasonably well with some but not all previous determinations.[13]

The application of LITPOMS not only recapitulates, via a single approach, the understanding of the calcium−calmodulin binding system gained over many decades but also provides new insights that cannot be easily accessed through other approaches. The outstanding spatial resolution of FPOP, as an example of MS-based structural proteomics, affords insights that previously required combinations of different approaches. The improving sensitivity of modern mass spectrometers allows these advances to be made with high picomoles of samples, whereas the irreversible labeling of FPOP broadens the application of LITPOMS by overcoming the major disadvantages of HD exchange,[25,36] making it possible to extend the former to the amino-acid residue level. With proper selection of labeling reagents,[37−39] it may be possible to characterize the affinity of a specific binding residue even at the nanomolar levels for many different ligands.[26] This bodes well for studies of signaling proteins,[3,4,40] where conformational changes are widespread over the protein and where concomitant changes in solvent accessibility are readily followed by reactions of free radicals and other reactive species.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.analchem.9b01062.

Detailed experimental conditions, global-level LITPOMS response, details about peptide selection, and an example of peptide-level data processing (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: mgross@wustl.edu.

ORCID

Xiaoran Roger Liu: 0000-0003-3198-1588
Mengru Mira Zhang: 0000-0001-9082-1272
Michael L. Gross: 0000-0003-1159-4636

Notes

The authors declare no competing financial interest.

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